

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Mechanical & Allied Engg. Semester: III

Subject Code & Name: BTMES304 Materials Science and Metallurgy

Max Marks: 60

Date:09-01-24

Duration: 3Hrs.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve *Any Two* of the following.

12

- A)** Sketch the following directions and planes within a cubic unit cell:

6

$[3\bar{1}2]$

$[10\bar{2}]$

$(2\bar{1}1)$

$(3\bar{1}2)$

- B)** Explain, in brief, the slip and twinning mechanisms of plastic deformation with the help of schematic diagrams.

6

- C)** Compare Brinell Hardness Test with Rockwell Hardness Test.

6

Q.2 Solve *Any Two* of the following.

12

- A)** Explain the Hume-Rothery's rules of solid solubility. Show the solidification of pure metals graphically.

6

- B)** What is lever rule? Derive the statement of Lever rule using schematic diagrams.

6

- C)** Discuss the eutectoid transformation on the equilibrium diagram of steels. Draw the microstructure of the eutectoid steel.

6

Q. 3 Solve *Any Two* of the following.

12

- A)** Explain the mechanism of removal of heat during cooling with the help of schematic diagrams/graphs. Give the requirements of an ideal quenchant.

6

- B)** Differentiate clearly between annealing and normalizing heat treatments. Show the temperature range of these treatments on the equilibrium diagram.

6

- C)** Describe the different types of flame hardening processes. What is the principle of induction hardening treatment?

6

Q.4 Solve <i>Any Two</i> of the following.	12
A) Compare macroscopy with microscopy w.r.t. definition, purposes, specimen preparation, merits , and demerits.	6
B) Define etching. Differentiate clearly between the mechanism of etching for single phase and two phase alloys.	6
C) Explain sulphur print test w.r.t. purpose, procedure, and chemical reactions involved.	6

Q. 5 Solve <i>Any Two</i> of the following.	12
A) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples	6
B) Discuss age/precipitation hardening process w.r.t. basic principle, conditions, and steps.	6
C) Describe magnetic particle inspection method using schematic diagrams.	6

***** End *****

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